WHITE PAPER EXCERPT

# Predicting High-Risk Non-Sufficient Funds

*Excerpt — Sections 1–2: Objective & Background*

**About this excerpt.** This is Sections 1 (Objective) and 2 (Background) from a longer internal research paper originally developed for a state lottery client, reproduced here to illustrate the business problem and operational context behind our analytical approach. Some figures, counts, and internal system names referenced elsewhere in the paper have been generalized or altered to protect client confidentiality — the context and reasoning are represented faithfully.

## 1. Objective

To research a methodology to predict and prevent the occurrence of high-risk non-sufficient funds instances to reduce risk and prevent loss for a Lottery and its retailers.

## 2. Background

A Lottery generates revenue by selling lottery games through its retailer base. Each week, Lottery retailers are invoiced, and the funds for payment are swept and remitted to the Lottery. When a retailer does not have adequate funds in their bank account for the weekly sweep, a Non-Sufficient Funds (NSF) instance occurs. The Lottery's internal bank processes weekly retailer account sweeps and notifies the Lottery via a file uploaded to Lottery databases that contains retailers that have incurred an NSF. In addition, the collection unit dispatches collectors that call retailers to inquire about the NSF. While a retailer has not paid their obligation on inventory, the Lottery will typically inactivate the retailer's Lottery terminal, temporarily halting any sales activity. The retailer may pay the NSF by check mailed to the Headquarters building or by paying the nearest District Office. In addition, collection letters are mailed to the retailer, and sales activity remains halted until the Lottery receives

compensation. The Lottery terminates business with the retailer after the fourth collection letter. Team members will generally retrieve all unsold Lottery products and Lottery equipment.

NSFs represent an infrequent occurrence at the Lottery, but a portion of the NSFs ultimately end up as lost revenue, accounting for losses of approximately $3.1 million from FY 2015–20. During that time, the Lottery conducted 6.2 million sweeps with NSFs occurring on 8,568 of those instances (.14% of the time). Out of the nearly $20.6 billion swept, $31.45 million returned as NSF (.15% of total dollars swept). The Lottery collected roughly 83.5% of the NSFs incurred in FY 2019–20.

Though NSFs occur infrequently and are largely collected on, understanding and predicting the outcome of an NSF does pose an opportunity to mitigate risk and prevent loss. Each NSF is a resource encumbrance when an NSF occurs. Staff key the NSF into the Lottery's financial system, and a collection analyst escalates the collections process by directly communicating with the retailer. A sales representative may need to visit the location. Further, when a retailer is inactive for any duration, they are not selling Lottery products. Retailers that NSF that fail to remit payment result in a loss to the Lottery, as the unpaid NSF eventually is processed as bad debt. Lastly, an NSF that leads to termination may represent a loss to the Lottery in future earnings.

## 2.1 Philosophy of an NSF

An NSF is an adverse occurrence for the Lottery and a retailer. However, NSFs, by nature, are not inherently wrong as they are a function of doing business at the Lottery. The way to reduce the NSF rate to zero is to terminate all retailers or generate no revenue, so the Lottery understands that an NSF rate is an accepted part of the Lottery business. The optimal goal is to a) understand an acceptable NSF rate for the Lottery to operate with (whether globally, by trade channel, by District Office, or some combination thereof), b) understand NSFs that are preventable (and NSFs that are not preventable), and c) understand and limit the occurrence of NSFs that may be detrimental for the Lottery.

NSFs occur for the following reasons, and an assessment of whether the NSF is preventable or not is listed below:

**Table 1: NSF Reasons and Preventability Table**

| NSF Type | Preventable |
|---|---|
| Retailers taking advantage of the NSF policy | No |
| Retailers committing fraud | Yes |
| Retailers improperly managing finances | No |
| Retailers incurring inventory saturation | Yes |
| Retailers experiencing long-term issues | Yes |
| Bank errors | No |

The Lottery cannot prevent retailers who NSF in all instances, as many instances are singular events. Further, the Lottery cannot prevent retailers who knowingly abuse the Lottery's NSF terms. Currently, retailers incur an NSF fee of $45 for each NSF, which has been treated as a "$45 short-term loan". Lastly, the Lottery cannot predict bank errors or frozen accounts. There will always be a challenge trying to build a predictive model to predict the occurrence of an NSF for all retailers as no internal data may lend to the predictability of singular instances from occurring.

The Lottery can work towards predicting some NSFs from occurring, however. Retailers who are potentially scratching their own instant ticket product and retailers experiencing inventory saturation may provide a signal. Lastly, retailers who experience long-term issues may be at higher risk for multiple NSFs or termination.

## 2.2 High-Risk NSFs

There are NSFs that are detrimental to the Lottery, even if an NSF, by nature, is not inherently bad. In most instances, the Lottery receives prompt payment for an NSF. However, when the NSF payment is not prompt, collections efforts require more Lottery resources. Further, these types of NSFs result in terminal deactivation and possibly termination. The NSF may eventually be accounted for as bad debt, an expense to the Lottery, and is ultimately treated as a loss.

The following table details the count, median NSF amount, and median days since the last NSF by days to pay.

**Table 2: NSF Details by Days to Pay**

| Name | Count | Med NSF | Med Days Since Last |
|---|---|---|---|
| NSF 0–10 days | 3,953 | $1,574 | 147 |
| NSF 11–20 days | 767 | $1,513 | 63 |
| NSF 21–30 days | 281 | $1,399 | 21 |
| NSF 31+ days | 1,368 | $2,331 | 21 |
| **Total** | **6,369** | **$1,689** | **42** |

NSF instances that are paid within 30 days or less share similar median amounts, however the days since the last NSF indicate that the longer it takes to pay an NSF, the higher probability the retailer may NSF again.

As NSFs are paid later, the risk of termination increases. The research team viewed unique retailers that NSFed compared to terminations that occurred within 60 days. The rate steadily increases the later NSFs are paid. When a retailer takes longer than 30 days, they experience a high likelihood of termination.

**Table 3: Termination Rates by NSF Days to Pay**

| Name | Retailer Count | Term Less Than 60 days | Percent |
|---|---|---|---|
| NSF 0–10 days | 2,027 | 162 | 7.99% |
| NSF 11–20 days | 586 | 81 | 13.82% |
| NSF 21–30 days | 217 | 50 | 23.04% |
| NSF 31+ days | 569 | 533 | 93.67% |
| **Total** | **3,399** | **826** | **24.30%** |

NSFs that are paid within 30 days indicate a retailer that still wants to be part of the Lottery family. Retailers who take 11–30 days to pay NSFs may be placed on probation, but they are still likely to remain a Lottery retailer. Retailers who take 31 or more days to pay their NSF are far less likely to remain within the Lottery retailer base.

The research team reviewed collection rates by days to pay. NSFs paid within 30 days were collected in full, while NSFs that took 31 or more days to pay were collected at only a 59% rate.

**Table 4: Collection Rates by NSF Days to Pay**

| Name | Total | Collected | Outstanding | Percent |
|---|---|---|---|---|
| NSF 0–10 days | $14,750,934 | $14,750,934 | $0 | 100% |
| NSF 11–20 days | $2,669,286 | $2,669,286 | $0 | 100% |
| NSF 21–30 days | $2,507,183 | $2,507,183 | $0 | 100% |
| NSF 31+ days | $6,277,467 | $3,726,679 | $2,550,788 | 59% |
| **Total** | **$26,204,870** | **$23,654,082** | **$2,550,788** | **90%** |

The research team categorized all NSFs taking 31 or more days to pay as a High-Risk Non-Sufficient Funds (HRNSF) instance. HRNSFs represent a major risk to the Lottery because they represent retailers at a higher risk of incurring another NSF relatively quickly, are likelier to lead to termination, and offer less chance of repayment.

## 2.3 Resource Allocation for NSFs

When a retailer incurs an NSF, the collections unit is responsible for getting in contact with them and getting a payment made as fast as possible. The process is performed by a series of collectors, and can be divided into 5 steps: populating the NSF list, contacting the retailer, keying in the NSF to the financial system, emailing the NSF list to Sales, and assisting the retailer with payment.

**Table 5: Resources Used by NSF Type**

| NSF Type | Populate List | Contact Retailer | Key in NSF | Email to Sales | Assist Retailer | Hours Used |
|---|---|---|---|---|---|---|
| NSF 0–10 days | 2 | 2.5 | 0.5 | 1 | 2 | 8 |
| NSF 11–20 days | 2 | 8.5 | 0.5 | 1 | 2 | 14 |
| NSF 21–30 days | 2 | 14.5 | 0.5 | 1 | 2 | 20 |
| NSF 31+ days | 2 | 20.5 | 0.5 | 1 | 2 | 26 |

As payments are prolonged, more resource time is used for each type of NSF. Each of the 4 collectors is paid an average of $8,618 a month, including salary and benefits. Based on an average of 176 working hours a month, this equates to $48.97 an hour.

**Table 6: Resource Cost by NSF Type**

| NSF Type | Hours | Rate | Total Resource |
|---|---|---|---|
| NSF 0–10 days | 8 | $48.97 | $391.76 |
| NSF 11–20 days | 14 | $48.97 | $685.58 |
| NSF 21–30 days | 20 | $48.97 | $979.40 |
| NSF 31+ days | 26 | $48.97 | $1,273.22 |

The longer it takes a retailer to pay their NSF, the more resource time and cost is allocated to the NSF instance. Further, resource time for each NSF may vary. Retailers may have multiple contacts with multiple phone numbers, and sometimes the contact information is incorrect or out of date. Lastly, retailers may need assistance with payment or understanding their invoice. The longer each NSF takes to be collected on, the more resource time is required to perform the contact attempts for each retailer. HRNSFs stress the time and cost of the collections unit to continually attempt contact with retailers for payment.

This excerpt covers only the Objective and Background sections of the full paper. The complete research paper also includes detailed methodology, findings on model performance and variable importance, fiscal impact analysis, and operational recommendations for deployment.